

Computação Gráfica / Computer Graphics

(L.EIC, 3rd course year)

Final project
Academic year 2025/2026

Version 1.0 draft
20th April 2026

1. General Objectives

This project aims to consolidate and extend the competence developed in the CGRA – Computer Graphics practical classes. Students will work in groups of three to build a complete 3D interactive scene from scratch, based on WebCGF, and by applying the techniques studied throughout the semester.

The main objectives concern:

- Apply geometric modelling techniques to create complex composite objects from simple primitives;
- Use geometric transformations (translation, rotation, scaling) and the composition of transformations to position, resize, shape, and to animate scene elements;
- Create and configure illumination, materials, and textures to achieve visually realistic surfaces;
- Develop custom GLSL shaders for procedural animation and visual effects;
- Implement real-time user interaction through keyboard controls and GUI elements;
- Animate scene components using time-dependent functions;
- Practice collaborative software development with version control (Git tags, commits, branching).

The theme of this year's scene is a **spring prairie landscape**, and your challenge is to build a game!

2. Introduction and gameplay

The scene is composed of a ground surface over an elevated terrain. On top, a cloudy sky. The ground contains multiple details and scattered elements. A covered light wagon (wagon, aka prairie schooner) and a barn are the two other relevant entities of the scene. Details for each of the previous assets are presented further in this document.

The game is a survival challenge: keep the wagon driver (you) “alive” for as much time as possible. In the end, the score is the total number of seconds of a single gameplay.

The barn is where the wagon starts, and it has a starting number of health points (HP). Health decreases over time and increases with food. Hay bales provide food by granting HPs when delivered to the barn. Hay bales appear randomly over the prairie.

The wagon needs to find and capture hay bales and bring them to the barn. Driving the wagon near a hay bale captures and puts it inside the wagon. When the wagon intersects a delimited circular area in front of the barn, the circular area boundary colour changes and the total amount of HPs from the captured hay bales is added to the wagon's HPs. At any given time, the schooner can only contain 2 hay bales. Hay bales are only visible when near them, but an animated arrow pinpoints them over the prairie.

Drive the wagon carefully along the path/fields, especially avoiding obstacles (e.g., rocks), as they can damage the wagon. To identify collisions, consider the distance between the centres of the objects. When smaller than a threshold, consider a collision to have occurred.

The challenge is to keep the wagon HP positive. If it reaches zero, the game is finished. As the economy of health points depends on several variables, the following “magic” numbers are for reference only and require tuning per project implementation:

- Wagon starts with an initial number of HPs (example: 100 HPs)
- Wagon loses HP/sec (example: 1 HP/sec)

- Each hay bale is worth several HPs (example: 50 HPs) when delivered to the barn.
- Wagon gets damage by hitting obstacles (example: loses a random HP number between 5 and 15)

3. Scene elements

Following is the description of scene elements:

Sky, clouds, and sun

Huge half-sphere geometry, faces turned inward, inverted normals, panoramic landscape environment.

Sun, Directional light/Spotlight

Advanced: Procedural clouds, animated cloud layer (as a second texture) near sky-sphere

Terrain elevation

Keep subtle rolling hills (use height maps for this), which are typical of the prairie.

Advanced: Terrain using procedural generation

Ground surface

Soil texture with dirt patches

Pathway for wagons

Scatter Elements

Rocks, multiple textures, and vertex perturbation

Advanced: Procedural generation of items

Flora

Parameter-based flowers. Parametrisation should foster natural randomness (textures, colours, scale, and flower placement around the terrain)

Advanced: procedural generation of several components of flora items (flowers)

Grass

Dense grass patches

Dead grass patches

Advanced: Shader to simulate “wind” over grass

After implementing terrain/ground, rocks, scatter elements, flowers, grass, and dry patches, the prairie field should have a natural look, as exemplified in Figure 1.



Figure 1: artistic illustrations of a prairie field (right side shows drier land).

Covered light wagon/prairie schooner

Create a hierarchical model of a wagon with the following mandatory components. See Figure 1 for illustration, but consider creating a geometrically simplified version. The basic components, from top to bottom, are: a cloth cover occupying half the wagon length (to see the hay bales), a wooden wagon bed, a front tongue to attach horses/mules, and 4 wheels.

Horses/mules should be imported in OBJ format. An extra WebCFG class is provided towards this purpose.

Advanced: Highly detailed (geometry and textures) wagon.



Figure 2: an illustrated perspective of a prairie schooner.

Wagon interaction mechanics

Smooth keyboard control of the front steering wheel to make the wagon turn Forward only. The wagon does not move backwards. Key map is:

- W accelerates forward (up to horse walk speed)
- A steers front wheels left
- D steers front wheels right
- S reduces speed
- P for hale pick-up
- L for hale drop

Advanced: acceleration using physics (kinematics).

Barn

Create a simple barn using a Cube with a wood plank texture.

Roof composed of half a prism with a dark wood texture.

Apply texture to mimic windows and doors.

Bale circular delimited area in front of the barn. The circular area changes look depending on whether the wagon intersects it.

Advanced: highly detailed barn and circular area, and visual feedback.

Interface elements

Use DAT.gui interface elements (numeric bars) to display:

- Current health points
- Instantaneous damage (HP) due to collision with rock/moving over rock
- Instantaneous health restored due to bay hale(s) reaching the barn
- Total number of bales dropped in the circular area in front of the barn
- Score (the time)

Animation

Animate the following parts:

- All the wheels of the Wagon rotate as the wagon advances.
- The front wheel axis is connected to the front tongue and should rotate together on a pivot point (in the centre of the axis) accordingly as the wagon turns left or right..
- Hay bale pinpointing arrow (vertical move)

Advanced: highly detailed wagon and horse movement.

Shaders

Apply shaders to the following scene elements:

- "Wind"-like grass moving
- Vertical pinpointing arrows

Advanced: visually convincing shaders.

4. Weekly plan and delivery deadline

The project spans 5+1 classes. From class 8 (starting 20th April) to class 12 (starting 25th May). The final version must be delivered in the Moodle "final project delivery" topic by the end of 29 May (Friday). The presentation of the final work is to be held in class 13 (starting 1st June)

The following calendar is a suggested schedule for work progress.

Semestre Week (Nº)	Tasks	Git Tag	# Screenshots Required
Week 8	Sky, sun, clouds	environment	2
Week 9	Terrain, ground, scatter elements	terrain	2
Week 10	Flowers and grass	flora	2
Week 11	Wagon + Interaction and animation	interaction	2
Week 12 Delivery preparation	Collectable system + UI.	gameplay	2
	README, video, final screenshots. (see section 6)	final	5 and 1 video
Week 13	Project presentation		

Git repository activity

As mentioned, you can adjust this calendar based on each group's progression strategy. However, you must push your work frequently to git to demonstrate progress. You can change the tag order depending on your group's progress strategy, but make sure all tags are pushed to the remote repository before the deadline.

6. Evaluation criteria

The top mark for each requirement corresponds to an optimal approach that cannot be further improved. No credit will be given for developments beyond what is specified, particularly to compensate for missing components. Creativity and originality should enhance – not replace – the required features.

Criterion	Weight
1. Explicit and procedural modelling of entities	4.0
2. Lighting conditions, materials, and textures	4.0
3. Overall scene and entities complexity	3.0
4. Animated entities	3.0
5. Advanced effects through shaders	3.0
6. Aesthetics and creativity ¹	2.0
7. Software Quality ²	1.0
TOTAL	20.0

1: Scene Aesthetics & Creativity evaluates how effectively the techniques from this project and prior practical assignments contribute to visual polish, originality, and overall scene coherence.

2: Software Quality considers code efficiency, modular structure, readability, and documentation (comments).

7. Final delivery

Make sure you submit all of the following artefacts via Moodle. Strictly follow the naming conventions. See the next section for details on the final presentation session.

README.md

A README.md file must be present in the repository root with the following content:

- Project title, class section, and group members (full name + student number)
- Brief description of the scene and features implemented
- Instructions to run the project (dependencies, how to launch)
- Keyboard controls reference table
- List of implemented features (indicating which bonus was chosen, if any)
- Known issues or limitations (if any)

- Screenshot thumbnails or links to the 5 required screenshots
- A declaration on AI use, to what extent and for what purpose.

Your source code project

Your entire project, containing the source code, the webCGF lib, and the updated README.md file. Everything is delivered in a single ZIP file. File naming convention: **project-t<class>g<group>-N.zip**. No other format allowed. Submit via the Moodle link “*Final project delivery*”.

Screenshots

Submit exactly 5 screenshots in Full HD resolution (1920×1080 pixels), in PNG format. File naming convention: **project-t<class>g<group>-N.png** (where N = 1..5). Submit at Moodle link “*Final images delivery*”

- Screenshot 1 — Overall scene overview (wide angle)
- Screenshot 2 — Flower rocks and floor detail
- Screenshot 3 — Wagon close-up (showing model detail, textures)
- Animated screenshot 4 — A Shader animation (**animated** GIF)

Demonstration video (1 minute)

Submit one video file (MP4 extension, H.264 codec) of exactly 1 minute. File naming: **project-t<class>g<group>.mp4**. The video must be a screen capture demonstrating:

- A tour of the full scene
- Wagon movement and animation (keyboard control, turning, acceleration)
- Pick-up and delivery of bay hales
- Shader(s)

Submit via the Moodle link “*Final video delivery*”. No other format is allowed except MP4.

8. Final presentation/question session

After delivery, each workgroup must present its work to the practical classes teacher (in the first week of June).